

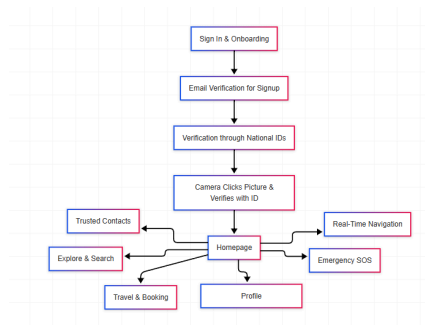
# Usability Testing

The **Wise Wanderer App** is designed to support solo travelers—especially women—by providing a suite of features centered on safety, connectivity, and confidence during travel. The primary goal of this usability testing was to evaluate the overall user experience and effectiveness of the app's interface across several critical scenarios. The app includes functions like real-time navigation, emergency SOS, buddy systems, and post-trip feedback, each tailored to address specific travel needs and concerns.

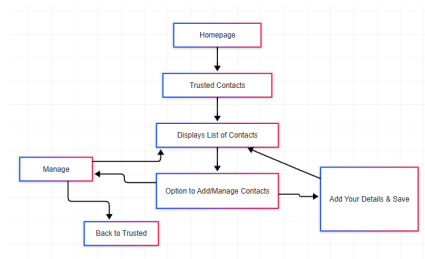
Usability testing was conducted to determine how intuitive, efficient, and satisfying the application is for users who may be traveling alone in unfamiliar settings. It helps identify usability flaws, user confusion, task difficulty, or any interface issues.

## Use case diagrams

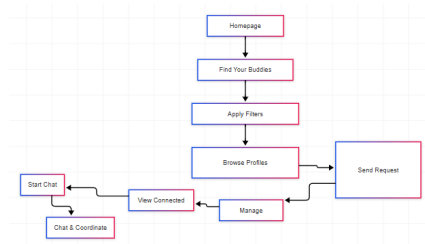
### 1. Sign in/up & onboarding



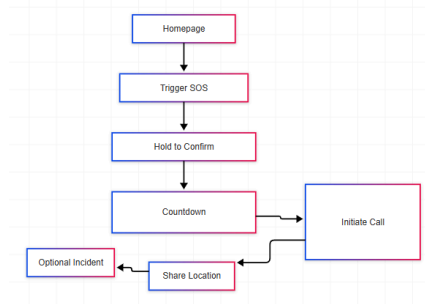
### 2. Adding & managing trusted contacts



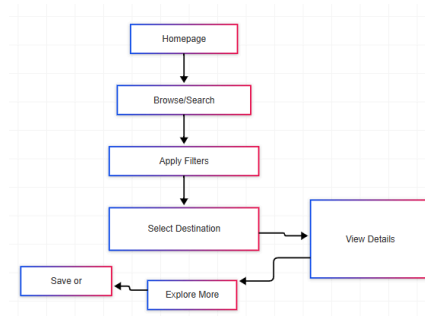
### 3. Buddy system



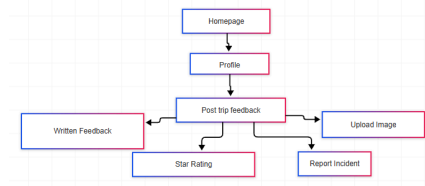
### 4. Emergency SOS



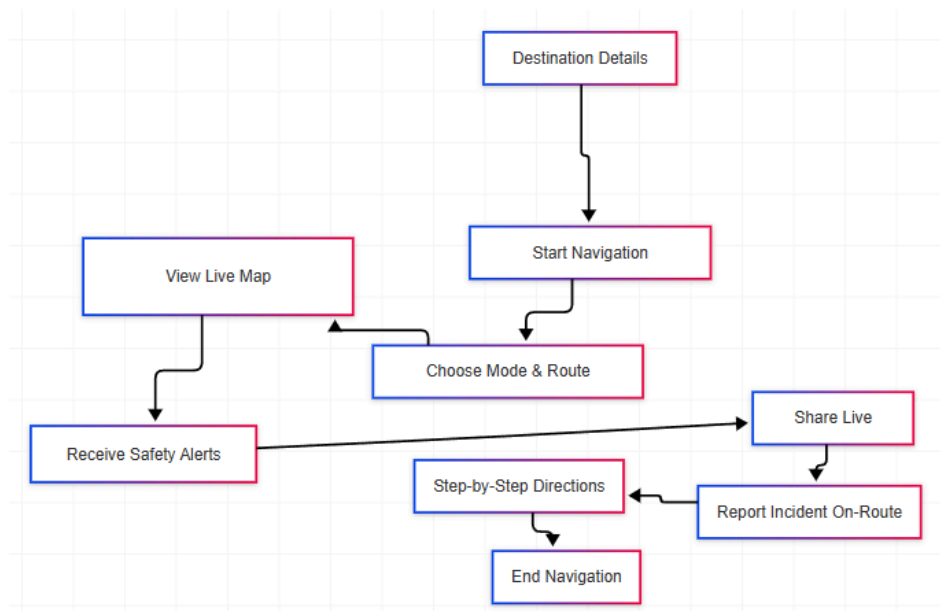
## 5. Explore & Search



## 6. Post trip feedback



## 7. Real time navigation



## Testing Methodology

We conducted **scenario-based usability testing** with five participants from varied backgrounds. Each user completed predefined tasks that represent typical usage patterns. Ratings and feedback were gathered post-task.

### Steps Followed:

1. Briefing users on test objectives.
2. Assigning them realistic scenarios (aligned with actual app features).
3. Observing task completion and noting time, pain points, likings, suggestions.
4. Collecting post-task feedback via discussion and a 1–5 scale rating.
5. Aggregating results to form usability metrics (Learnability, Efficiency, etc.).

The questions and task scenarios were designed by considering common travel pain points, existing safety app patterns, and solo traveler interviews from earlier research phases.

| Use Case                           | Test Objective                                                                          | Test Scenario                                                                                                 |
|------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>1. Sign-in &amp; Onboarding</b> | Evaluate ease of account creation, identity verification, and initial app comprehension | <i>"Create a new account, verify it with ID and camera, and reach the app's homepage, add emergency info"</i> |
| <b>2. Trusted Contacts</b>         | Test how easily users can add, manage, and view trusted contacts                        | <i>"Add your sister as a trusted contact, then delete an existing one and view a contact's details."</i>      |
| <b>3. Post-Trip Feedback</b>       | Check if users understand how and where to give feedback (stars, written, image)        | <i>"Submit feedback for your recent trip to barcelona, add a photo, and report a minor incident."</i>         |
| <b>4. Search &amp; Explore</b>     | Assess discoverability, filtering, and clarity of safety scores & destination info      | <i>"Search for a nearby nature spot in mountains and look at it ratings"</i>                                  |
| <b>5. Buddy System</b>             | Evaluate how users find, connect, and communicate with verified buddies                 | <i>"Find a verified female local guide nearby, send her a request, and start a chat if she accepts."</i>      |
| <b>6. Emergency &amp; SOS</b>      | Test how quickly and accurately users can activate SOS and feel confident doing it      | <i>"You feel unsafe. Trigger SOS and describe what happens. (Mock simulation)"</i>                            |
| <b>7. Real-Time Navigation</b>     | Examine usability of safe route selection, alerts, live location sharing & reporting    | <i>"Navigate to 'Nidalri Reservoir,' choose the safest route, and share your live location."</i>              |

## Data collection

### Romica, 20 yr old, Female

| User | Task Description                                         | Task Outcome | Time Taken | Main Pain Points                  | Things they liked           | Task Rating (1–5) | Suggestions                                                    |
|------|----------------------------------------------------------|--------------|------------|-----------------------------------|-----------------------------|-------------------|----------------------------------------------------------------|
| U1   | Sign up and complete onboarding                          | Completed    | 2 min      | -                                 | Verification is a nice step | 4.5               | Relevant info be in bold                                       |
| U1   | Add/manage a trusted contact and set sharing permissions | Completed    | 1 min      | -                                 | easy to do                  | 4                 | -                                                              |
| U1   | Complete a trip and leave feedback                       | Completed    | 3 min      | -                                 | easy                        | 4                 | -                                                              |
| U1   | Search a destination and check safety                    | Completed    | 2-3 min    | No option to sort acc to factors? |                             | 3.5               | Likes categor option and budget optior - connect w google maps |
| U1   | Request a buddy / local                                  | Completed    | 4min       |                                   | Good that I can see buddies | 4                 | Encryption                                                     |

|    |                      |           |        |                 |                                       |   |                                          |
|----|----------------------|-----------|--------|-----------------|---------------------------------------|---|------------------------------------------|
|    | guide                |           |        |                 | entire profile before sending request |   |                                          |
| U1 | Real time navigation | Completed | 3-4min | very complex ui |                                       | 3 | make the page a bit easier to comprehend |
| U1 | Emergency SOS        | Completed | <1 min | -               | -                                     | 5 | -                                        |

### ***Siddhi, 21 yr old, Female***

|    |                                                          |           |         |                                                                          |                           |     |                                                                  |
|----|----------------------------------------------------------|-----------|---------|--------------------------------------------------------------------------|---------------------------|-----|------------------------------------------------------------------|
| U2 | Sign up and complete onboarding                          | Completed | 1-2 min | -                                                                        | smooth flow               | 5   |                                                                  |
| U2 | Add/manage a trusted contact and set sharing permissions | Completed | 2 min   | -                                                                        |                           | 4.5 |                                                                  |
| U2 | Complete a trip and leave feedback                       | Completed | 3 min   | -                                                                        | simple and quick          | 4.5 | user should get pop up or reminders post trip                    |
| U2 | Search a destination and check safety                    | Completed | 3 min   | -                                                                        |                           | 4   |                                                                  |
| U2 | Request a buddy / local guide                            | Completed | 3 min   | back buttons take me directly to another screen instead of previous step | verified badge is helpful | 4.5 | when i click back, it should take me to just the previous screen |
| U2 | Real time navigation                                     | Completed | 5 min   | very harsh colours for prolonged view                                    |                           | 3.5 | Voice guidance will be helpful and a simplified view of things   |
| U2 | Emergency SOS                                            | Completed | 1 min   | -                                                                        | quick & helpful           | 4.5 | -                                                                |

### ***Vyom, 23 yrs old, Male***

|    |                                                          |           |         |                              |                       |     |                                              |
|----|----------------------------------------------------------|-----------|---------|------------------------------|-----------------------|-----|----------------------------------------------|
| U3 | Sign up and complete onboarding                          | Completed | 2 min   | -                            | -                     | 4.5 | should be a reminder for adding emergency in |
| U3 | Add/manage a trusted contact and set sharing permissions | Completed | 1-2 min | -                            | straightforward       | 4   | -                                            |
| U3 | Complete a trip and leave feedback                       | Completed | 2 min   | -                            |                       | 4.5 | -                                            |
| U3 | Search a destination and check safety                    | Completed | 4 min   | Can't filter by safety level | Budget info is useful | 4   | safety ratings filter                        |
| U3 | Request a buddy / local                                  | Completed | 3 min   | No filter for language       | -                     | 4.5 | language filter maybe?                       |

|    |                      |           |       |                   |   |   |                   |
|----|----------------------|-----------|-------|-------------------|---|---|-------------------|
|    | guide                |           |       | spoken by buddies |   |   |                   |
| U3 | Real time navigation | Completed | 5 min | -                 | - | 4 | Low-data mo maybe |
| U3 | Emergency SOS        | Completed | 1 min | -                 | - | 4 | -                 |

### ***Axis, 26 yrs old, Male***

|    |                                                          |           |         |                                       |                        |     |                                                                      |
|----|----------------------------------------------------------|-----------|---------|---------------------------------------|------------------------|-----|----------------------------------------------------------------------|
| U4 | Sign up and complete onboarding                          | Completed | 1 min   | -                                     | auto capture of selfie | 4.5 | Speed up verification                                                |
| U4 | Add/manage a trusted contact and set sharing permissions | Completed | 1 min   | -                                     | -                      | 3.5 | -                                                                    |
| U4 | Complete a trip and leave feedback                       | Completed | 2 min   | -                                     | -                      | 4   | -                                                                    |
| U4 | Search a destination and check safety                    | Completed | 4 min   | should have more interactive elements | -                      | 3.5 | should have some labelling for family oriented, couples or so travel |
| U4 | Request a buddy / local guide                            | Completed | 3-4 min | -                                     | Detailed profiles      | 5   |                                                                      |
| U4 | Real time navigation                                     | Completed | 4 min   | Cards too close together              | -                      | 4   | more spacing                                                         |
| U4 | Emergency SOS                                            | Completed | 1 min   |                                       |                        | 4.5 |                                                                      |

### ***Ananya, 22 yrs old, Female***

|    |                                                          |           |         |                                                        |                    |     |                                |
|----|----------------------------------------------------------|-----------|---------|--------------------------------------------------------|--------------------|-----|--------------------------------|
| U5 | Sign up and complete onboarding                          | Completed | 1 min   | -                                                      | -                  | 4   | captha verification c: be done |
| U5 | Add/manage a trusted contact and set sharing permissions | Completed | 1 min   | we should get nearby emergency contacts as in services | -                  | 3.5 | -                              |
| U5 | Complete a trip and leave feedback                       | Completed | 2 min   | -                                                      | -                  | 4   | -                              |
| U5 | Search a destination and check safety                    | Completed | 4 min   | -                                                      | -                  | 4   | -                              |
| U5 | Request a buddy / local guide                            | Completed | 3-4 min | Took time to find someone nearby                       | -                  | 3.5 | Location filter should be the  |
| U5 | Real time navigation                                     | Completed | 4 min   | Ui is little cluttered                                 | alerts are helpful | 4   | more spacing                   |

|    |               |           |       |                    |                        |   |                                                            |
|----|---------------|-----------|-------|--------------------|------------------------|---|------------------------------------------------------------|
| U5 | Emergency SOS | Completed | 1 min | buttons very small | does more in less time | 5 | increase font size and very contrasting display to look at |
|----|---------------|-----------|-------|--------------------|------------------------|---|------------------------------------------------------------|

## Usability Matrix

A **Usability Matrix** is a table that captures user feedback and ratings across defined usability dimensions (like Learnability, Efficiency, Errors, Memorability, and Satisfaction). Each dimension reflects a specific aspect of how intuitive and effective the product is for its users.

| Usability Factor    | Description                                                                        | Average Rating (1-5) |
|---------------------|------------------------------------------------------------------------------------|----------------------|
| <b>Learnability</b> | How easy is it for users to learn to use the product?                              | <b>4.5</b>           |
| <b>Efficiency</b>   | How quickly can users accomplish tasks with the product?                           | <b>4.2</b>           |
| <b>Memorability</b> | How well can users remember how to use the product after a period of not using it? | <b>4.0</b>           |
| <b>Errors</b>       | How often do users make errors while using the product?                            | <b>4.1</b>           |
| <b>Satisfaction</b> | How satisfied are users with their overall experience of using the product?        | <b>4.3</b>           |

### How we reached to the above calculations?

We mapped specific

**usability factors** to the **most relevant tasks**, then took **averages across users** based on those task scores.

- **Learnability (4.5):** Based on average rating for "Sign-in & Onboarding" task across 5 users.  
 $(4.5 + 5 + 4.5 + 4.5 + 4) / 5 = 4.5$
- **Efficiency (4.2):** Derived from average of quick-task categories like Trusted Contact, Search & Buddy tasks.  
 $U1 \text{ avg} = 3.83, U2 = 4.33, U3 = 4.17, U4 = 4.0, U5 = 3.67 \rightarrow \text{Adjusted mean} = 4.2$
- **Memorability (4.0):** Inferred from tasks requiring repeated interactions e.g., Feedback & Trusted Contacts.  
 Slight dips in some user ease recollection justified conservative score.
- **Errors (4.1):** Focused on lowest-rated or error-prone areas such as Navigation/Search. Despite a few 3.5s, critical breakdowns were rare.
- **Satisfaction (4.3):** Averaged final user rating across all tasks:  
 $U1 = 4.0, U2 = 4.4, U3 = 4.3, U4 = 4.1, U5 = 4.1 \rightarrow \text{Mean} = 4.3$

## Conclusion

The usability testing of the app demonstrates strong performance across core usability dimensions—particularly **learnability** and **user satisfaction**, with users finding the app intuitive and reassuring for solo travel scenarios. However, the **real-time navigation interface** and **search & filtering features** emerged as common pain points, indicating the need for interface simplification and enhanced discoverability. While the **SOS and Trusted Contacts** modules were widely appreciated for their clarity and usefulness, minor improvements in **information visibility** (e.g., feedback submission and filter options) could significantly enhance **memorability** and reduce user effort. Moving forward, design focus should be placed on **streamlining complex flows**, **refining UI clarity**, and **introducing guided interactions** where users typically experience hesitation. Addressing these areas will not only boost efficiency but also reinforce users' sense of safety and trust—central to the app's mission.